

Renormalization-Group Invariant Parity-Doublet Model for Nuclear and Neutron-Star Matter

Mattia Recchi,¹ Lorenz von Smekal,^{1,2} and Jochen Wambach^{3,4}

¹*Institut für Theoretische Physik, Justus-Liebig-Universität, Heinrich-Buff-Ring 16, 35392 Giessen, Germany*

²*Helmholtz Forschungsakademie Hessen für FAIR (HFHF), GSI Helmholtzzentrum für Schwerionenforschung, Campus Giessen*

³*Technische Universität Darmstadt, 64289, Darmstadt, Germany*

⁴*University of Illinois at Urbana-Champaign, Urbana, IL 61801, USA*

(Dated: November 11, 2025)

The Parity-Doublet Model (PDM) is a chirally invariant effective theory for strong-interaction matter involving nucleons and their opposite-parity partners in a parity-doubling framework. We introduce a multiplicatively renormalizable mean-field approach to include the baryonic vacuum contributions to the resulting grand-canonical potential in an explicitly renormalization-group invariant form. As an application, we evaluate the pertinent thermodynamics of two-flavor symmetric and asymmetric nuclear matter, focusing on the restoration of spontaneously broken chiral symmetry at baryon densities and temperatures relevant for the astrophysics of neutron stars. Special attention is paid to the effect of the baryonic vacuum fluctuations on the evolution of chiral condensate with baryon density and temperature for specific choices of the chirally invariant baryon mass m_0 to demonstrate the importance of consistently including these vacuum fluctuations in the PDM.

I. INTRODUCTION

Understanding the structure of nuclear matter under extreme conditions is key to strong-interaction physics with applications in astrophysics and high-energy nuclear physics. As first-principles lattice-QCD (LQCD) calculations cannot be applied in the relevant regions of the phase diagram because of the fermion sign problem, modeling chiral-symmetry restoration through effective chiral theories is a central objective for the description of dense and hot baryonic matter realized in core-collapse supernovae, the interior of neutron stars (NS), in NS mergers, and in heavy-ion collisions at center-of-mass energies of a few GeV.

Chiral models, which have a long history [1], are based on the empirical observation that hadronic parity partners have different masses as a consequence of spontaneously broken chiral symmetry in the non-perturbative QCD vacuum. They reproduce the (nearly) exact degeneracy of the spectral properties of chiral partners of opposite parity in the chirally restored phase as a general consequence of chiral invariance of the QCD Lagrangian for massless quarks. But there are also parity partners that are no chiral partners, i.e. which are not directly related by chiral symmetry. This opens interesting possibilities for effective Lagrangians with chiral symmetry, especially for baryons.

For high baryon density, the parity-doublet model [2], as a generalization of the linear sigma model [3] and a chirally symmetric version of the Walecka model [4], is particularly interesting. Although respecting chiral symmetry it involves massive baryonic parity partners with a common finite mass m_0 and only the mass splitting is generated by the chiral condensate. The mass m_0 , the origin of which remains unspecified, serves a parameter of the model. The Parity-Doublet Model (PDM) offers the intriguing possibility of a chirally restored phase with

massive parity-doubled baryons [5, 6]. This phenomenon has been confirmed for baryon octet and decuplet screening masses in finite-temperature LQCD simulations [7]. In recent years, the PDM has gained attention for its application to nuclear saturation properties, the description of chiral phase transitions in nuclear matter, and the behavior of the chiral condensate at finite density and in isolated NS [8–10]. Furthermore, extensions to flavor $SU(3)$ [11–13] have been made to study the role of strangeness in NS. There are also interesting consequences for the experimental heavy-ion programs at FAIR/GSI energies [14, 15].

The work described here builds on and extends previous two-flavor studies [10, 16] in a one-loop renormalized mean-field (MF) approximation. It presents a novel multiplicatively renormalizable treatment of the fermionic vacuum fluctuations, ensuring explicit renormalization group (RG) invariance of the grand-canonical potential. This procedure has proven quantitatively successful for pure-isospin QCD matter, where a direct comparison of the analogous RG-invariant MF calculation in the quark-meson model with LQCD results is possible [17]. In particular, the RG-invariant scale parameter Λ , here of the PDM, is generated by dimensional transmutation, as usual, to replace the renormalization scale and the relevant dimensionless interaction strength without introducing a new parameter.

The paper is organized as follows: In Sec. II we recall the basic ideas of the PDM, especially emphasizing the role of the chirally invariant mass term m_0 . After introducing the model for the two-flavor case, we discuss the resulting grand-canonical potential and the RG-invariant treatment of the divergent fermionic vacuum contribution (VC). We then present the strategy for fixing the model parameters through vacuum phenomenology and well-established properties of low-density symmetric and asymmetric nuclear matter. Sec. III is devoted to selected results on the phase structure of the chiral condensate as

a function of the thermodynamic variables, temperature T , baryo-chemical potential μ_B , and isospin-chemical potential μ_I for a range of chiral mass parameters m_0 , contrasting the results with and without including the VC. Our particular focus is on the evolution of the chiral condensate in the vicinity of the liquid-gas (LG) transition of symmetric nuclear matter. We then turn the astrophysical applications for β -equilibrated NS matter by discussing the composition, the Equation of State (EoS), and the resulting Mass-Radius (M-R)-relations for evolved NS together with their tidal deformabilities, as well as the thermal index which is relevant for proto-NS evolution and binary NS mergers. Sec. IV summarizes the pertinent findings of our paper and discusses future directions.

II. THE PARITY-DOUBLET MODEL

In this section we briefly review the main ideas of the two-flavor PDM, establishing the notation and discussing the resulting grand-canonical potential $\Omega(T, \mu_B, \mu_I)$. We then focus in detail on the RG-invariant treatment of the divergent fermionic VC to Ω . The rest of the section is devoted to the choice of parameters and the fitting procedure.¹

A. The Model

In its two-flavor version, the PDM is a generalization of the linear sigma model [3] in which the sigma-meson and the pion fields are coupled to the nucleon $N(939)$ and its negative-parity partner, here assumed to be the $N^*(1535)$ resonance. One starts with two sets of spin-1/2 baryons in two-flavor chiral $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ representations, N_1 and N_2 , with chirally invariant Yukawa couplings to the $O(4)$ vector $(\sigma, \vec{\pi})$ of sigma-meson and pion fields. Requiring so-called “mirror assignment” [18], where the left-handed N_2 transforms as the right-handed N_1 and vice versa under $SU(2)_L \times SU(2)_R$, it is then possible to introduce a chirally invariant fermion mass m_0 , common to both parity-partner baryons, to model purely gluonic mass contributions from the QCD scale anomaly [19, 20] which do not rely on chiral symmetry breaking. The fermionic part $\mathcal{L}_F$ of the (Euclidean) Lagrangian is then given by

$$\begin{aligned} \mathcal{L}_F = & \bar{N}_1 (\not{\partial} + g_1 (\sigma + i\gamma_5 \vec{\tau} \cdot \vec{\pi})) N_1 \\ & + \bar{N}_2 (\not{\partial} + g_2 (\sigma - i\gamma_5 \vec{\tau} \cdot \vec{\pi})) N_2 \\ & + m_0 (\bar{N}_1 \gamma_5 N_2 - \bar{N}_2 \gamma_5 N_1) \end{aligned} \quad (1)$$

with the Yukawa couplings g_1 and g_2 , where the opposite signs in the (pseudo-scalar) pion-baryon couplings reflect

the mirror assignment of the chiral transformations of N_1 and N_2 . With spontaneous chiral symmetry breaking, the σ field receives a non-vanishing vacuum expectation value $\langle \sigma \rangle \equiv \sigma_0 = f_\pi \approx 92$ MeV, leading to additional Dirac mass contributions $m_1 = g_1 f_\pi$ and $m_2 = g_2 f_\pi$ to N_1 and N_2 . The physical positive $N_+ = N(939)$ and negative $N_- = N^*(1535)$ parity baryon fields are then obtained from diagonalizing the mass matrix in the Nambu-Gorkov space of parity-partner baryons, so that chiral-symmetry breaking is responsible for the mass splitting of the two.

For space-time independent, constant meson fields the diagonalization is straightforward, yielding

$$\mathcal{L}_F = \bar{N}_+ (\not{\partial} + m_+) N_+ + \bar{N}_- (\not{\partial} + m_-) N_- , \quad (2)$$

with the physical masses $m_\pm$ given by

$$m_\pm(\phi) = \frac{1}{2} \sqrt{(g_1 + g_2)^2 \phi^2 + 4m_0^2} \pm \frac{g_1 - g_2}{2} \phi , \quad (3)$$

in terms of the constant radial $O(4)$ -field variable $\phi^2 = \sigma^2 + \vec{\pi}^2$. Of course, with the correct vacuum alignment, we may set $\phi \rightarrow \sigma$, but the form in Eq. (3) explicitly reflects the fact that the fermion determinant is chirally invariant.

In the vacuum, with $\sigma_0 = f_\pi$, we use $m_N \equiv m_+(f_\pi) = 939$ MeV for the nucleon and $m_{N^*} \equiv m_-(f_\pi) = 1510$ MeV for the $N^*(1535)$ as its parity partner [21]. Since for any given value of m_0 :

$$\begin{aligned} g_2 - g_1 &= \frac{m_{N^*} - m_N}{f_\pi} \approx 6.2 , \\ g_1 + g_2 &= \frac{\sqrt{(m_N + m_{N^*})^2 - 4m_0^2}}{f_\pi} , \end{aligned} \quad (4)$$

this then uniquely fixes g_1 and g_2 . As chiral symmetry gets restored in the medium, the sigma-field diminishes, either in a crossover transition or a first(second)-order phase transition. When chiral symmetry is fully restored ($\sigma = 0$), according to Eq. (3), the parity-partner baryons become degenerate in mass with finite m_0 .

In the mesonic sector, the MF approximation essentially amounts to adding an effective potential $U(\sigma, \vec{\pi})$ to the Lagrangian, which is of the form

$$U(\sigma, \vec{\pi}) = V(\phi^2) - c\sigma , \quad (5)$$

with a polynomial in the $O(4)$ -invariant ϕ^2 , conventionally starting with

$$V(\phi^2) = -\frac{m^2}{2} \phi^2 + \frac{\lambda}{4} \phi^4 + \dots . \quad (6)$$

In the present work, we will use an 8th-order polynomial for $V(\phi^2)$ of equivalent form but with slightly unusual coefficients, as we will detail below.

The term $-c\sigma$ accounts for the explicit breaking of chiral symmetry due to finite u and d quark masses. In terms of the pion decay constant f_π and the pion mass

¹ We will use natural units $\hbar = c = k_B = 1$ throughout.

m_π , we have $c = f_\pi m_\pi^2$. In the chirally broken vacuum only the σ field is non-vanishing and equal to f_π . We will further assume that there is no pion condensation in nuclear and neutron matter and hence the classical pion field vanishes. Thus the σ -field is the only chiral order parameter throughout. While the σ -field is treated classically, we will include fermionic vacuum fluctuations, as detailed in the next subsection, in a renormalized MF-approximation [10, 22, 23] which is equivalent to integrating the purely fermionic functional RG flow [24]. As we will see below, the inclusion of these fluctuations is crucial for the evolution of the σ -field and hence the restoration of chiral symmetry at high density.

B. Grand-Canonical Potential

In the path-integral representation of the grand-canonical partition function one includes chemical potentials μ_B for baryon number and μ_I for isospin imbalance (here chosen to be positive for neutron excess) by adding $-(\mu_B - \mu_I \tau^3) \gamma_0$ to the diagonal blocks of the Nambu-Gorkov Dirac operator in Eq. (1). We will also include the short-range repulsion between baryons in the form of four-Fermi interactions in the isoscalar and (neutral) isovector vector channels. Assuming equal strengths for both,² N and N^* , this is achieved by including corresponding Hubbard fields ω_μ and ρ_μ in the diagonal blocks as well. The repulsive nature implies pure imaginary cou-

plings, replacing

$$\not{\phi} \rightarrow \not{\phi} - i\gamma^\mu (g_\omega \omega_\mu - g_\rho \rho_\mu \tau^3) \quad (7)$$

in Eqs. (1) and/or (2), together with additional quadratic terms in the mesonic potential, defining

$$U(\sigma, \vec{\pi}, \omega, \rho) = V(\phi^2) - c\sigma + \frac{m_\omega^2}{2} \omega^2 + \frac{m_\rho^2}{2} \rho^2. \quad (8)$$

The strengths of the repulsive interactions are solely determined by the ratios g_ω^2/m_ω^2 and g_ρ^2/m_ρ^2 , so that the assignment of the mass parameters is arbitrary. For comparison with conventional practice, we use the physical vector-meson masses, $m_\omega = 783$ MeV and $m_\rho = 776$ MeV, here. The MF approximation for the Hubbard fields then implies pure imaginary saddle points in their zero components, $\omega_0 = i\bar{\omega}$ and $\rho_0 = i\bar{\rho}$ with positive constant $\bar{\omega}$, $\bar{\rho} \geq 0$.³ In the Nambu-Gorkov Dirac operator we can include these potentially non-vanishing expectation values by simple shifts in the corresponding chemical potentials, defining

$$\tilde{\mu}_\tau = \mu_B - g_\omega \bar{\omega} - \tau(\mu_I - g_\rho \bar{\rho}), \quad (9)$$

where we have introduced the coefficient $\tau = \pm 1$, with $\tau = +1$ for the proton and its charged parity partner and $\tau = -1$ for the neutron and the neutral N^* , corresponding to the eigenvalues of the Pauli isospin matrices.

With these conventions, the grand-canonical MF potential of the two-flavor PDM can easily be derived to be of the form

$$\Omega(T, \mu_B, \mu_I) = -2T \sum_{\tau=\pm 1} \sum_{\pm} \int \frac{d^3 p}{(2\pi)^3} \left(\ln \cosh \left(\frac{E_{\pm} - \tilde{\mu}_\tau}{2T} \right) + \ln \cosh \left(\frac{E_{\pm} + \tilde{\mu}_\tau}{2T} \right) \right) + U(\sigma, \vec{\pi}, i\bar{\omega}, i\bar{\rho}), \quad (10)$$

where $E_{\pm} \equiv \sqrt{p^2 + m_{\pm}^2(\phi)}$ are the meson-field dependent single-particle energies of N and N^* , for which we can identify $\phi = \sigma$ for vanishing pion field.

Focusing on the fermionic part of the potential we note that the momentum integration in Eq. (10) is ultraviolet divergent. We can separate finite parts from the logarithmically divergent ones, which are contained solely in the VC denoted by Ω_{vac} . First we replace $\cosh(x)$ with $\cosh(|x|)$, in order to explicitly take out the $T \rightarrow 0^+$ limit

of the fermionic potential:

$$\ln \cosh \left(\frac{|E - \tilde{\mu}_\tau|}{2T} \right) = \frac{|E - \tilde{\mu}_\tau|}{2T} - \ln \left(1 - f \left(\frac{|E - \tilde{\mu}_\tau|}{T} \right) \right), \quad (11)$$

with the Fermi-Dirac distribution $f(x) = 1/(e^x + 1)$.

Inserting Eq. (11) into the grand-canonical potential, we obtain:

$$\Omega = \Omega_T + \Omega_n + \Omega_{\text{vac}} + U, \quad (12)$$

² Chiral symmetry and parity then uniquely define the two allowed four-fermion interactions in these channels [25].

³ It amounts to a stationary-phase approximation with complex saddle points, from where the stable Lefschetz thimbles emerge parallel to the real axis.

with

$$\Omega_T = 2T \sum_{\tau=\pm 1} \sum_{\pm} \int \frac{d^3 p}{(2\pi)^3} \left(\ln \left(1 - f \left(\frac{|E_{\pm} - \tilde{\mu}_{\tau}|}{T} \right) \right) + \ln \left(1 - f \left(\frac{|E_{\pm} + \tilde{\mu}_{\tau}|}{T} \right) \right) \right), \quad (13)$$

$$\Omega_n = 2 \sum_{\tau=\pm 1} \sum_{\pm} \int \frac{d^3 p}{(2\pi)^3} (E_{\pm} - |\tilde{\mu}_{\tau}|) \theta(|\tilde{\mu}_{\tau}| - E_{\pm}), \quad (14)$$

$$\Omega_{\text{vac}} = -4 \sum_{\pm} \int \frac{d^3 p}{(2\pi)^3} \sqrt{p^2 + m_{\pm}^2(\sigma)}. \quad (15)$$

Thus Ω separates into a thermal part Ω_T depending on temperature and chemical potential, which vanishes for $T = 0$, a density-dependent zero-temperature contribution Ω_n from the Fermi sea, which vanishes for $\tilde{\mu}_{\tau} = 0$, and Ω_{vac} , the VC containing the ultraviolet divergent parts, which needs to be regularized and renormalized as described in the next subsection.

Note that Ω is invariant under $\tilde{\mu}_{\tau} \rightarrow -\tilde{\mu}_{\tau}$. This implies that it is only a function of $|\tilde{\mu}_{\tau}|$. In the following, we consider $\tilde{\mu}_{\tau} > 0$ without any loss of generality.

The Landau free-energy density as a function of T , μ_B and μ_I is obtained from the MF grand-canonical potential upon extremizing Ω with respect to the constant meson and Hubbard fields, i.e. solving

$$\frac{\partial \Omega}{\partial \sigma} = 0, \quad \frac{\partial \Omega}{\partial \bar{\omega}} = 0, \quad \frac{\partial \Omega}{\partial \bar{\rho}} = 0, \quad (16)$$

at fixed T , μ_B and μ_I . The ultraviolet divergent contribution from Ω_{vac} , which is often omitted in MF calculations (in the so-called “no-sea approximation”), is field dependent and thus relevant in the equation for σ .

C. Vacuum Contribution

We now analyze the VC denoted by Ω_{vac} . Regularization and renormalization follow standard procedures [10, 22, 23]. Here we adopt an RG invariant formulation analogous to the one used in Ref. [17] for the quark-meson model which provides an accurate description of the current LQCD data at vanishing μ_B but finite isospin density. Being a variant of the RG-MF formulation used previously, for convenience, we only briefly describe the main steps, adapted to the PDM where necessary.

Firstly, as frequently also used for Casimir forces, the ultraviolet divergent integral in Eq. (15) can be regularized analytically with

$$\int p^2 dp (p^2 + m^2)^{(1-\epsilon)/2} = \frac{m^{4-\epsilon} \sqrt{\pi} \Gamma(\frac{\epsilon-4}{2})}{4\Gamma(\frac{\epsilon-1}{2})}. \quad (17)$$

For $\epsilon \rightarrow 0^+$ this yields:

$$\Omega_{\text{vac}} = \frac{1}{\pi^2} \sum_{\pm} \left(\frac{m_{\pm}^4}{4\epsilon} - \frac{m_{\pm}^4}{16} \left(1 - 4 \ln 2 + 4 \ln(m_{\pm}/\nu) \right) + \mathcal{O}(\epsilon) \right), \quad (18)$$

where $m_{\pm} \equiv m_{\pm}(\sigma)$, and ν is the renormalization scale. We see that the logarithmically divergent terms are proportional to

$$\frac{1}{4} (m_+^4 + m_-^4) = \frac{m_0^4}{2} + (g_1^2 + g_2^2 - g_1 g_2) m_0^2 \phi^2 + \frac{1}{4} (g_1^4 + g_2^4) \phi^4. \quad (19)$$

This implies that we can adopt the standard MF renormalization of the celebrated Gell-Mann-Levy model by introducing counter-terms of precisely this same structure, differing from the quartic meson self-interactions $\propto \phi^4$ only by the term $\propto m_0^2 \phi^2$ and an irrelevant (field-independent) constant. We therefore *define* the relevant terms of the tree-level mesonic potential for a multiplicatively renormalizable MF theory of the PDM by

$$V(\phi^2) \equiv -\frac{1}{2} \frac{m^2}{g_1^2 + g_2^2} (m_+^2(\phi) + m_-^2(\phi)) + \frac{\lambda}{4(g_1^4 + g_2^4)} (m_+^4(\phi) + m_-^4(\phi)). \quad (20)$$

Since

$$\frac{1}{2} (m_+^2 + m_-^2) = m_0^2 + \frac{1}{2} (g_1^2 + g_2^2) \phi^2, \quad (21)$$

the first term in Eq. (20) yields the same quadratic term as in Eq. (6). Therefore Eq. (20) only differs from the conventional form in Eq. (6) by the additional term $\propto m_0^2 \phi^2$ from Eq. (19).

For stability at large fields, we can simply add irrelevant terms of the (RG-invariant) form

$$V_6(\phi^2) \equiv \frac{c_6}{6m_0^2} (m_+^6(\phi) + m_-^6(\phi)), \quad (22)$$

$$V_8(\phi^2) \equiv \frac{c_8}{8m_0^4} (m_+^8(\phi) + m_-^8(\phi)), \quad (23)$$

with some dimensionless constants $c_6, c_8 > 0$. For large fields, these behave as

$$V_6(\phi^2) \sim \frac{c_6}{6m_0^2} (g_1^6 + g_2^6) \phi^6, \quad (24)$$

$$V_8(\phi^2) \sim \frac{c_8}{8m_0^4} (g_1^8 + g_2^8) \phi^8. \quad (25)$$

Expressing the potential in this way, we can absorb the logarithmic divergences by a renormalization of the effective coupling

$$u \equiv \frac{g_1^4 + g_2^4}{\lambda}, \quad (26)$$

which controls the MF approximation in the PDM, in a way analogous to g^4/λ in the procedure of Ref. [17] for the quark-meson model. This is achieved by noting that the pre-factors of all terms proportional to $m_\pm^4/4$ in Ω can be combined to yield,

$$\frac{1}{u} + \frac{1}{\pi^2} \left(\frac{1}{\epsilon} + \ln 2 - \frac{1}{4} - \ln(m_\pm/\nu) \right) \equiv \quad (27)$$

$$\frac{1}{u_R(\nu)} - \frac{1}{\pi^2} \ln(m_\pm/\nu),$$

where $u_R(\nu) = Z_u^{-1}u$ is the renormalized effective coupling at the renormalization scale ν , and

$$Z_u = 1 + \frac{u}{\pi^2} \left(\frac{1}{\epsilon} + \ln 2 - \frac{1}{4} \right). \quad (28)$$

Note that, in absence of pion condensation for sufficiently small μ_I , which is the case considered here, it is in principle sufficient to renormalize the quartic meson-coupling, $\lambda_R = Z_\lambda^{-1}\lambda$ with $Z_\lambda^{-1} = Z_u$. Due to the need of meson-field renormalization in the seagull terms, in the presence of pion condensation, this is not sufficient in general, however. As in Ref. [17], the complete multiplicative MF renormalization, is then defined by $\phi_R = Z_\phi^{-1/2}\phi$ for the meson fields, $m_R^2 = Z_m^{-1}m^2$ for their mass parameter, and with equal renormalization factors $g_{1,R}^2 = Z_g^{-1}g_1^2$, $g_{2,R}^2 = Z_g^{-1}g_2^2$, for both Yukawa couplings. One can then conveniently choose $Z_u = Z_g^2/Z_\lambda$ and Z_g as the two independent renormalization factors that are needed, with the others determined by $Z_\phi Z_g = Z_\phi Z_m = 1$ [17]. Matching powers of meson fields and Yukawa couplings, such as $(g_1^2 + g_2^2)\phi^2$, are then RG-invariant, and so is the

ratio

$$\tilde{m}^2 \equiv \frac{m^2}{g_1^2 + g_2^2} \quad (29)$$

which determines the mass term in the mesonic potential in Eq. (20). With the RG-invariant mass parameter m_0 of the PDM, the physical baryon masses $m_\pm(\sigma)$ are then automatically RG invariant as well, as they must.

The total effective potential for the sigma- and pion fields, including the VC, can then be collected into

$$V_{\text{eff}}(\phi) = \sum_{\pm} \left\{ -\tilde{m}^2 \frac{m_\pm^2(\phi)}{2} + \left(\frac{1}{u_R(\nu)} - \frac{1}{\pi^2} \ln \frac{m_\pm(\phi)}{\nu} \right) \frac{m_\pm^4(\phi)}{4} + c_6 \frac{m_\pm^6(\phi)}{6m_0^2} + c_8 \frac{m_\pm^8(\phi)}{8m_0^4} \right\} \quad (30)$$

which is clearly RG invariant with $\beta(u) = u^2/\pi^2$. Since it is renormalized and finite, we drop the index R again, denoting by $u \equiv u_R(\nu)$ the scale-dependent renormalized effective coupling from now on. Introducing the scale parameter Λ of the RG-invariant MF model via $\Lambda \equiv \nu e^{\pi^2/u}$, we can explicitly eliminate the remaining RG-scale ν dependence in Eq. (30) by the replacement

$$\frac{1}{u} - \frac{1}{\pi^2} \ln \frac{m_\pm(\phi)}{\nu} = -\frac{1}{\pi^2} \ln \frac{m_\pm(\phi)}{\Lambda}. \quad (31)$$

Alternatively, we may introduce the effective coupling $u_0 = u(\nu_0)$, using the chirally-invariant PDM mass m_0 as the reference scale, $\nu_0 = m_0$, to define our renormalization scheme. This then leads to an explicitly RG-scale independent form of the grand-canonical potential Ω as follows:

$$\Omega(T, \mu_B, \mu_I) = \sum_{\pm} \left\{ -\tilde{m}^2 \frac{m_\pm^2(\phi)}{2} + \frac{1}{\pi^2} \left(\ln \frac{\Lambda}{m_0} - \ln \frac{m_\pm(\phi)}{m_0} \right) \frac{m_\pm^4(\phi)}{4} + c_6 \frac{m_\pm^6(\phi)}{6m_0^2} + c_8 \frac{m_\pm^8(\phi)}{8m_0^4} \right\}$$

$$- c\sigma - \frac{m_\omega^2}{2} \bar{\omega}^2 - \frac{m_\rho^2}{2} \bar{\rho}^2 + \frac{2}{\pi^2} \sum_{\tau=\pm 1} \sum_{\pm} \int p^2 dp (E_\pm - |\tilde{\mu}_\tau|) \theta(|\tilde{\mu}_\tau| - E_\pm)$$

$$+ \frac{2T}{\pi^2} \sum_{\tau=\pm 1} \sum_{\pm} \int p^2 dp \ln \left(1 - f \left(\frac{|E_\pm - \tilde{\mu}_\tau|}{T} \right) \right) + \ln \left(1 - f \left(\frac{|E_\pm + \tilde{\mu}_\tau|}{T} \right) \right). \quad (32)$$

Since, in this scheme, we have $m_0 = \Lambda e^{-\pi^2/u_0}$, we need $\Lambda \gg m_0$ for any reasonably small and positive u_0 (choosing $\Lambda = m_+(f_\pi) = m_N$ would correspond to the scheme used in Ref. [10]). Here, we will use the parameters $\tilde{m}^2 = m^2/(g_1^2 + g_2^2)$ and $\tilde{\lambda}_0 = 1/u_0 = \ln(\Lambda/m_0)/\pi^2$ to fix $\sigma_0 = f_\pi$ (at the reference scale) and the σ -meson

(curvature) mass parameter m_σ , see below.

D. Parameter Fixing and Best Fit

We now focus on the choice of the parameters for the PDM. These are fixed both from hadron phenomenology and the saturation properties of nuclear matter.

The distinctive parameter of the PDM is the chirally invariant mass m_0 which enters the physical masses of the parity partners from Eq. (3), with

$$m_+(f_\pi) = m_N, \quad m_-(f_\pi) = m_{N^*}, \quad (33)$$

where for any given m_0 the two Yukawa couplings g_1 and g_2 are chosen such that the vacuum masses of the nucleon and the $N^*(1535)$ are given by $m_N = 939$ MeV and $m_{N^*} = 1510$ MeV as described in Sec. II A. The value of the chirally invariant baryon mass m_0 is rather poorly constrained, however. On physical grounds, we expect it to vary roughly between

$$500 \text{ MeV} \leq m_0 \leq 860 \text{ MeV}. \quad (34)$$

The upper value results from the QCD trace anomaly in the chiral limit of vanishing light-quark masses [19, 20], while m_0 between 500 MeV and 550 MeV has been estimated from QCD sum rules [26].

To bracket these possibilities we chose a range of m_0 between 500 MeV and 800 MeV in the following. For any fixed value of m_0 the parameters in the baryonic Lagrangian (1) are thus fixed from Eqs. (4).

The leading mesonic couplings are m^2 and λ , or equivalently, $\tilde{m}^2$ and u as defined in Eqs. (29) and (26). While $\tilde{m}^2$ is RG invariant, the latter is RG-scale ν dependent. Instead of $u \equiv u(\nu)$, we therefore use the RG invariant scale parameter $\Lambda = \nu e^{\pi^2/u} = m_0 e^{\pi^2/u_0}$ in our scheme, as introduced above. These two are fixed in the vacuum, requiring the expectation value of the σ -field and its Euclidean (curvature) mass parameter at the reference scale⁴ to be given by f_π and m_σ^2 ,

$$\begin{aligned} \left. \frac{d\Omega}{d\sigma} \right|_{\phi=(f_\pi, \vec{0})} &= 0, \\ \left. \frac{d^2\Omega}{d\sigma^2} \right|_{\phi=(f_\pi, \vec{0})} &= m_\sigma^2. \end{aligned} \quad (35)$$

Since the mass parameter of the σ -meson as a broad two-pion resonance, is not very well constrained phenomenologically, we allow it to vary between 400 MeV and 550 MeV in the fitting procedure described next.

In addition to Λ and $\tilde{m}^2$, the remaining parameters that need to be fixed are given by the dimensionless mesonic c_6 and c_8 in (30) and the strengths g_ω and g_ρ of isoscalar and isovector vector repulsion between the

baryons. To achieve this we use a best-fit procedure by minimizing the residual function

$$\text{Res}(\vec{x}) = \sum_i \frac{(x_i^{\text{exp}} - x_i)^2}{\sigma_i^2}, \quad (36)$$

where $\vec{x}^{\text{exp}}$ is the vector of phenomenological quantities with associated uncertainties $\vec{\sigma}$. The minimization is performed through a global-local hybrid approach: we use the DIRECT algorithm to search for a region containing the global minimum and then refine the solution using a standard Levenberg–Marquardt algorithm.

The quantities that enter in the fit are the nuclear saturation density, n_0 , the binding energy per nucleon of symmetric nuclear matter, E_B , its compressibility at the saturation point, K_∞ , the symmetry energy, E_{sym} , and the value of the in-medium chiral condensate $\sigma(n_0)$, deduced from other sources.

Following the recent analysis in [27], n_0 and E_B are taken as

$$\begin{cases} n_0 = 0.157 \pm 0.010 \text{ fm}^{-3}, \\ E_B = 15.97 \pm 0.40 \text{ MeV}. \end{cases} \quad (37)$$

The compressibility is defined as

$$K_\infty = 9n_0^2 \left. \frac{\partial^2 E}{\partial n^2} \right|_{n_0}, \quad (38)$$

where E is the energy per particle of the system. The value of K_∞ is around 240 ± 20 MeV [28].

To fix the value of g_ρ we need a quantity related to the isospin asymmetry parameter $\delta = (n_n - n_p)/(n_p + n_n)$ where n_n and n_p are the neutron and proton density, respectively. The symmetry energy is then given by

$$E_{\text{sym}} = \left. \frac{1}{2} \frac{\partial^2 E}{\partial \delta^2} \right|_{n_0, \delta=0}. \quad (39)$$

We take E_{sym} as 31 ± 2 MeV [29].

Since one of the main focuses of our study is the dynamics of the in-medium chiral condensate, we have decided to include in the parameter-fixing procedure also a quantity related to that, namely the value of the chiral condensate at saturation density. A natural way to introduce such a constraint is through the nucleon sigma term, σ_N , which measures the explicit chiral symmetry breaking contribution to the nucleon mass. In particular, σ_N is defined as the nucleon matrix element of the light quark mass term,

$$\sigma_N = m_q \langle N | \bar{u}u + \bar{d}d | N \rangle = m_q \frac{\partial m_N}{\partial m_q}, \quad (40)$$

and provides a measure of the sensitivity of the nucleon mass to changes in the quark masses. It also governs the leading linear density dependence of the in-medium condensate. In fact, using the Feynman-Hellman theorem and the Gell-Mann-Oakes-Renner relation, one can

⁴ In general, these relations involve finite renormalization factors from meson field renormalization.

relate the reduction of the condensate at finite density to the value of σ_N , leading to the expression:

$$\sigma(n_0) = f_\pi \left(1 - \frac{\sigma_N n_0}{f_\pi^2 m_\pi^2} \right), \quad (41)$$

where n_0 denotes the saturation density and f_π , m_π are the pion decay constant and mass, respectively.

The physical value of σ_N is still somewhat controversial, with lattice simulations providing values roughly around 44 ± 4 MeV [30, 31], while the analysis of πN -scattering yields a value of about 56 ± 4 MeV, without isospin-breaking effects [32, 33] (also see the compilation in Ref. [34] which reports a value between the two).

Moreover, it is well established that these vacuum values lead to an unrealistically strong in-medium reduction of the chiral condensate once higher-order corrections are considered. The saturation density might not be small enough for the linear approximation in (41) to be valid. To include these higher-order effects, we follow Ref. [35] which can be used to provide a suitable *effective in-medium sigma term* at saturation density of $\tilde{\sigma}_N(n_0) = 35 \pm 8$ MeV to incorporate the non-linear corrections to Eq. (41), where the central value is based on $\sigma_N = 45$ MeV, and hence consistent within current errors with the recent lattice estimates of the standard nucleon-sigma term at vanishing density.

Plugging $\tilde{\sigma}_N(n_0)$ into Eq. (41) (in the place of σ_N) we obtain:

$$\sigma(n_0) = 68 \pm 15 \text{ MeV}. \quad (42)$$

The best-fit parameters are shown in Tab. I together with the resulting σ -meson mass parameter m_σ . The parameter sets are labeled by the chiral baryon mass parameter m_0 used on input. The second column shows the RG-invariant scale parameter Λ implicitly defined by the inverse of the quartic meson coupling λ at the reference scale $\nu = m_0$ in our scheme as explained above. In the no-sea approximation (labeled “No Vacuum Contribution”) the effective coupling $u = (g_1^4 + g_2^4)/\lambda$ is of course not scale dependent, and we simply define $\Lambda \equiv m_0 e^{\pi^2/u}$ as a proxy for λ for a better comparison in this case. The third column lists the meson mass parameter $\tilde{m}^2 = m^2/(g_1^2 + g_2^2)$ in units of Λ . These two are essentially determined from f_π (not shown in the table) and m_σ . The remaining dimensionless parameters c_6 , c_8 , g_ω and g_ρ of Tab. I then lead to the physical quantities listed in Tab. II.

The temperature T_c of the critical point of the LG phase transition and the symmetry-energy slope parameter,

$$L = 3n_0 \left. \frac{\partial E_{\text{sym}}}{\partial n} \right|_{n_0}, \quad (43)$$

were not included in the fit procedure, but are listed as predictions of the model in the last two columns of Tab. II for reference. The current empirical value for T_c

is around 17 MeV but the experimental determination of this quantity is quite challenging and the uncertainty is more than 10 % [36]. As shown in the table, values of m_0 between 600 MeV and 800 MeV are favored by this quantity. The slope parameter L is also marked by relatively large uncertainties, with an accepted empirical value around 60 ± 10 MeV [37, 38]. More recent studies based on astrophysical observations are consistent within present errors but tend towards somewhat lower values [39, 40]. The comparatively large values obtained in the present study (which are in line with Ref. [16]), on the other hand, might indicate the need to include a $\rho - \omega$ interaction term [41] in the future.

III. RESULTS

A. Phase Structure

In Fig. 1 we display the phase diagram of the model for different values of m_0 , without (left column) and with (right column) the VC. We report the values of the in-medium chiral condensate as a function of temperature and baryon chemical potential for $m_0 = 500$ MeV, 600 MeV and 800 MeV, respectively. Without the VC we obtain a first-order chiral transition in all cases (red lines). For $m_0 = 500$ MeV the first-order line extends to $\mu_B = 0$.⁵ For the higher m_0 values the temperature of the chiral critical point is around that of the LG transition or even below (e.g., around 15 MeV for $m_0 = 800$ MeV).

With the inclusion of the VC the picture changes drastically, especially for the smaller values of m_0 . The transition is pushed to much higher chemical potentials and becomes a smooth crossover for most values of m_0 we have considered, including $m_0 = 700$ MeV (not shown). Only for the largest $m_0 = 800$ MeV, where the effect of the VC is comparatively small, the chiral first-order transition persists, although at a somewhat higher μ_B and with a somewhat smaller critical temperature than without the VC. Overall, the results we obtain for $m_0 = 800$ MeV are consistent with those of Refs. [10, 16].

Currently, there is no experimentally conclusive direct evidence for a first-order chiral phase transition in the QCD phase diagram although the existence of a chiral critical point is expected from state-of-the-art QCD calculations with functional methods [45, 46] and a majority of QCD-based models. Since in our study the chiral transition is located in very high-density regions of the phase diagrams (at densities $n_\chi \sim 10 n_0$), we expect the details of the hard-core repulsion of the nuclear interaction to

⁵ We note that the parameter set associated to $m_0 = 500$ MeV and no VC causes sizable difficulties in the numerical solutions. It is conceivable that forcing the system to reproduce the observables at low values of m_0 drives the system into a non-physical minimum of the residual function in Eq. (36).

No Vacuum Contribution							
m_0 [MeV]	Λ [MeV]	$\tilde{m}^2/\Lambda^2$	c_6	c_8	g_ω	g_ρ	m_σ [MeV]
500	515	0.909×10^{-2}	-3.054×10^{-4}	1.046×10^{-5}	8.28	4.06	528.9
600	620	0.686×10^{-2}	-4.843×10^{-4}	2.386×10^{-5}	7.44	4.12	486.1
700	731	0.618×10^{-2}	-9.110×10^{-4}	6.412×10^{-5}	6.35	4.18	440.3
800	881	0.848×10^{-2}	-2.936×10^{-3}	2.956×10^{-4}	4.84	4.24	418.4

Vacuum Contribution							
m_0 [MeV]	Λ [MeV]	$\tilde{m}^2/\Lambda^2$	c_6	c_8	g_ω	g_ρ	m_σ [MeV]
500	1358	1.656×10^{-2}	0.682×10^{-2}	-1.520×10^{-4}	7.03	4.04	475.0
600	1389	1.658×10^{-2}	0.941×10^{-2}	-2.911×10^{-4}	6.51	4.09	451.0
700	1422	1.661×10^{-2}	1.225×10^{-2}	-4.963×10^{-4}	5.91	4.19	400.0
800	1513	1.646×10^{-2}	1.398×10^{-2}	-6.339×10^{-4}	4.76	4.10	400.0

TABLE I. Model parameters obtained from fitting the phenomenological properties of nuclear matter in Tab. II without (top) and with the VC (bottom) for different values of fixed m_0 , from left: the scale parameter Λ (without the VC simply defined as $\Lambda = m_0 \exp\{\pi^2 \lambda / (g_1^4 + g_2^4)\}$), the mass parameter $\tilde{m}^2 = m^2 / (g_1^2 + g_2^2)$ in the mesonic potential (20), and the dimensionless parameters c_6 , c_8 , g_ω , g_ρ for the strengths of higher-order couplings and vector repulsion (m_σ is allowed to vary between 400 MeV and 550 MeV, but otherwise not constrained and quoted for reference here, see text).

No Vacuum Contribution							
m_0 [MeV]	E_B [MeV]	n_0 [fm $^{-3}$]	E_{sym} [MeV]	K_∞ [MeV]	$\sigma(n_0)$ [MeV]	L [MeV]	T_c [MeV]
500	16.0	0.157	31.0	240	67.19	84	17.2
600	16.0	0.157	31.0	240	67.19	83	17.3
700	16.0	0.157	31.0	240	67.19	82	18.0
800	16.0	0.157	31.0	240	67.19	80	21.7

Vacuum Contribution							
m_0 [MeV]	E_B [MeV]	n_0 [fm $^{-3}$]	E_{sym} [MeV]	K_∞ [MeV]	$\sigma(n_0)$ [MeV]	L [MeV]	T_c [MeV]
500	15.9	0.162	30.99	244	72.13	82	14.9
600	15.8	0.161	31.01	244	71.21	82	15.5
700	16.0	0.157	30.95	240	69.49	81	16.5
800	15.8	0.164	31.01	259	66.73	80	21.5

TABLE II. Phenomenological properties of nuclear matter near saturation as obtained from the best-fit parameters in Tab. I without (top) and with the VC (bottom) for matching values of m_0 , from left: the binding energy E_B per nucleon of symmetric nuclear matter, its saturation density n_0 , the symmetry energy E_{sym} , the compressibility K_∞ of infinite nuclear matter, and the in-medium chiral condensate $\sigma(n_0)$ at saturation density. The slope parameter L and the LG critical temperature T_c are not used in the best-fit procedure, but are quoted here as resulting model predictions.

play a crucial role.⁶

In Fig. 2 we summarize the evolution of the zero-temperature chiral condensate for the m_0 values between 500 MeV and 800 MeV, with the parameters of Tab. I, for symmetric nuclear matter (left panel) and pure neutron matter (right panel). The latter is relevant for evolved isolated NS, to be discussed below.

For isospin symmetric matter in the left panel the first jump is located at $m_N - E_B \approx 923$ MeV and is associated with the onset of self-bound nuclear matter.⁷ The

dashed lines show a second jump corresponding to a first-order chiral phase transition. The position of this jump strongly depends on the value of m_0 and is particularly low for small values of m_0 . The inclusion of the VC makes the transition a smooth crossover, except for the largest $m_0 = 800$ MeV. In this case the effect of the VC is much less dramatic and affects only the position of chiral transition and the size of the jump that is reduced instead of being eliminated. Importantly, however, we note that reducing the chiral baryon mass m_0 from 800 MeV down to 500 MeV with or without the VC has the opposite effect on the position of the chiral transition, where the m_0 dependence is stronger without the VC. In particular, a low-density chiral transition is only possible for small chiral baryon mass m_0 in combination with the no-sea approximation, i.e. without the VC. We therefore conclude that it is an artifact of this rather crude approximation entirely neglecting the VC.

In the right panel of Fig. 2 there is no first jump but a smooth onset of repulsive pure neutron matter placed at $\mu_n = m_N$, where $\mu_n = \mu_B + \mu_I$ is the chemical potential

⁶ Implicitly, using the PDM assumes that baryonic degrees of freedom survive chiral restoration. However, the densities found in the present study could possibly be high enough to make a transition to deconfined quark matter which, of course, cannot be described by a hadronic model such as the PDM.

⁷ We do not study the mechanically unstable region (e.g., low-density phase separation) of nuclear matter but only investigate stable homogeneous matter. In particular we solve all equations at fixed μ_B which means that densities below n_0 cannot be explored for symmetric nuclear matter.

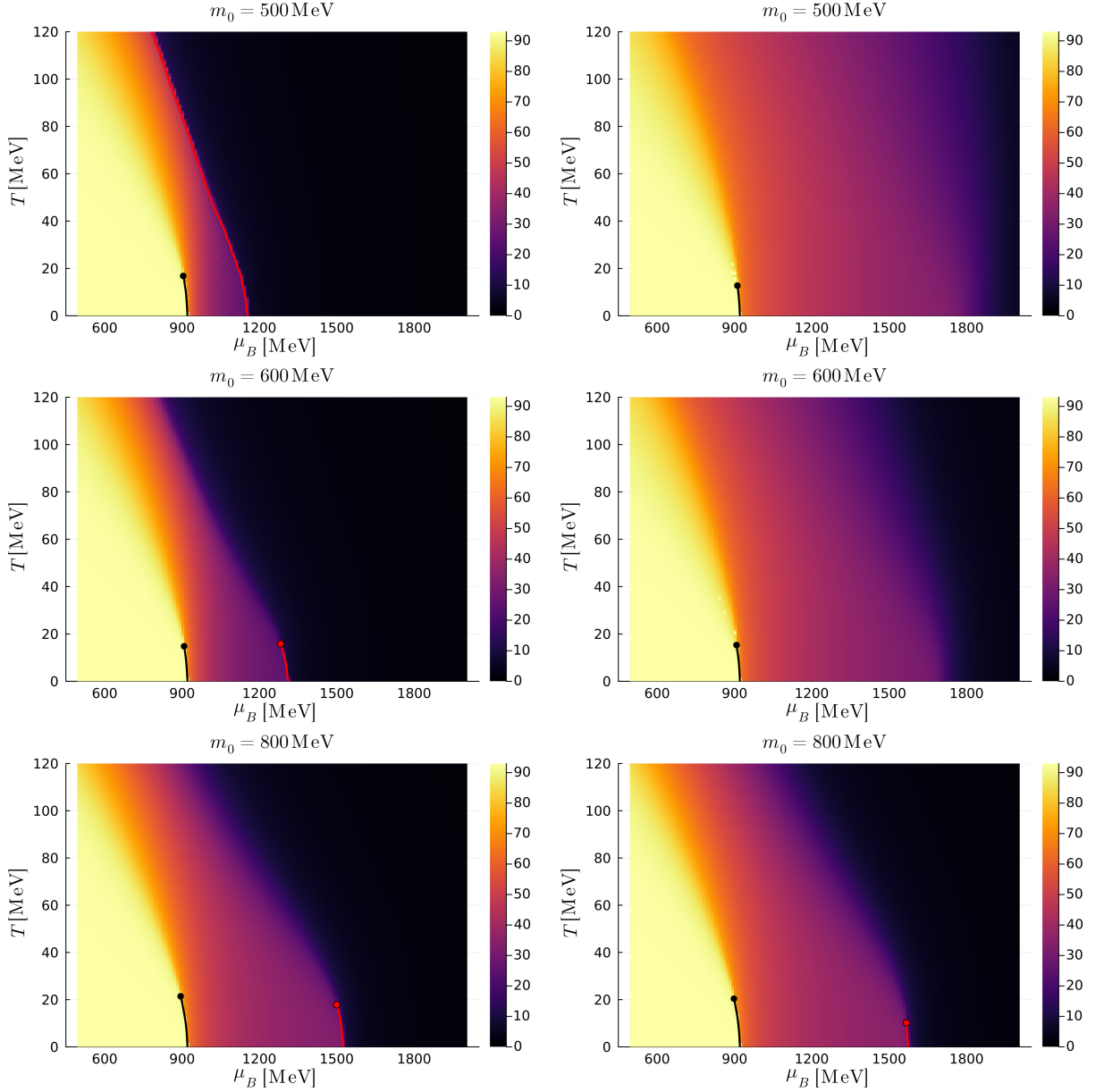


FIG. 1. Phase diagrams for symmetric nuclear matter, where the heatmaps display the value of the in-medium chiral condensate $\sigma(T, \mu_B)$ in the (T, μ_B) -plane. The left column does not include the VC while the right one does. The black lines indicate the LG transition while the red lines mark the chiral transition.

for neutrons with our conventions for μ_I (as introducing an excess of neutrons). The conclusions on the m_0 dependence of the chiral transition are qualitatively the same as for symmetric nuclear matter.

Fig. 3 shows the value of the isospin asymmetry parameter $\delta = (n_n - n_p)/n$ as a function of μ_B and μ_I at vanishing temperature, i.e. the zero-temperature phase diagram. We choose δ because it highlights the various phases which are characterized by different neutron and proton fractions. The phase diagram is displayed up to $\mu_I = 70 \text{ MeV} \approx m_\pi/2 \text{ MeV}$ because at isospin chemical potential $\mu_I = m_\pi/2$ one expects the onset of

pion condensation that we do not consider in this work but refer to the analogous discussion in the quark-meson model [17, 47]. Negative values of μ_I produce the same zero-temperature phase diagram with $\delta \rightarrow -\delta$, we can therefore limit our discussion to $\mu_I \geq 0$ (here for neutron excess) without loss.

The lower, empty part of the figure, represents the zero-density (Silver-Blaze) region associated with the vacuum of the theory. Following the μ_B axis at $\mu_I = 0$ one sees the onset of self-bound symmetric nuclear matter at $\mu_B = m_N - E_B \approx 923 \text{ MeV}$. The step between the white vacuum region and the dark symmetric matter

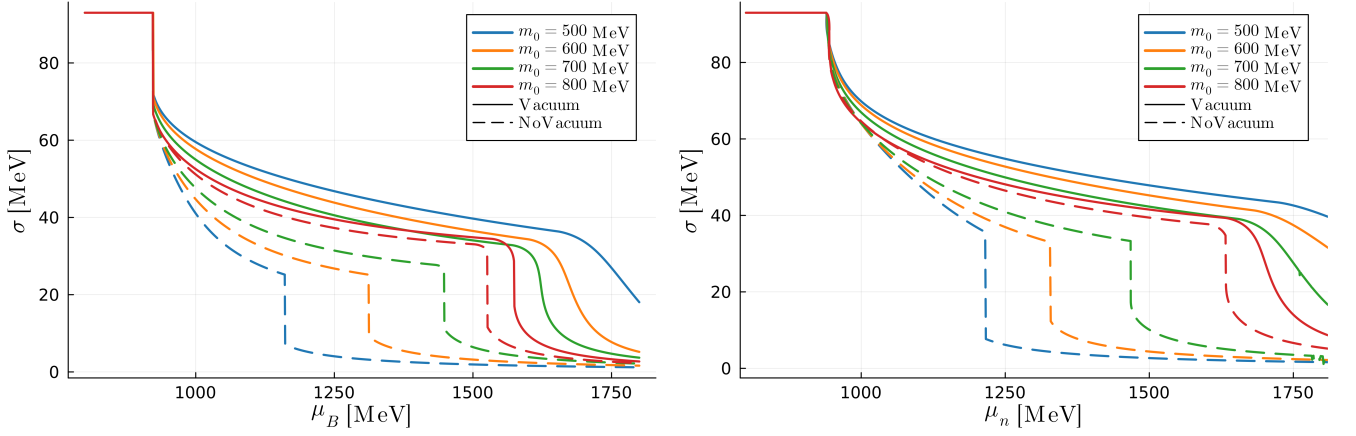


FIG. 2. The chiral condensate at $T = 0$ for symmetric nuclear matter as a function of $\mu_B = (\mu_p + \mu_n)/2$ (left panel) and pure neutron matter as a function of the neutron chemical potential $\mu_n = \mu_B + \mu_I$ (right panel) for different values of m_0 . The full lines denote the results with inclusion of the VC while for the dashed lines we leave out the VC. In the left panel, the jump from the vacuum value of $\sigma = f_\pi$ at $\mu_B = m_N - E_B$ represents the onset of self-bound nuclear matter.

region ($\delta = 1$) is discontinuous in density with a jump from $n = 0$ to $n_0 \approx 0.16 \text{ fm}^{-3}$. For non-vanishing but small $0 < \mu_I \lesssim 20 \text{ MeV}$ the red line indicated in Fig. 3 separates predominantly symmetric nuclear matter (with a small isospin asymmetry) discontinuously from the vacuum. At $\mu_I \approx 20 \text{ MeV}$, the red line intersects with the black line along $\mu_B = m_N - \mu_I$ which marks the energy cost to produce a single unbound neutron (neutron drip). Beyond this point the red line separates predominantly symmetric nuclear matter from essentially pure neutron matter. The separation is discontinuous in density up to the red critical point at around $\mu_I \approx 36 \text{ MeV}$ where the discontinuity vanishes in a continuous second-order transition. Beyond this critical point in the zero-temperature phase diagram the LG transition turns into a smooth crossover. This critical point is the limit $T_c(\mu_I) \rightarrow 0$ of

the LG critical point starting at $T_c \approx 17 \text{ MeV}$ for symmetric nuclear matter at $\mu_I = 0$.

The bright yellow region above the $\mu_B = m_N - \mu_I$ line is the phase of pure neutron matter ($\delta = 1$). Since neutron matter is not self-bound, the transition between vacuum and the yellow region, across the black line, is continuous in density. The top part of the diagram, above the yellow region, is characterized by arbitrary mixtures of neutrons and protons in asymmetric nuclear matter. The light-blue curve indicates mixtures of neutrons and protons that satisfy the conditions of β -equilibrium in neutron star matter, to be discussed in the next section.

The qualitative aspects of the zero-temperature phase diagram shown in Fig. 3 are model-independent features of QCD. As such they are reproduced by the PDM, here exemplified with $m_0 = 700 \text{ MeV}$, but essentially unchanged with the other parameter sets of Tab. I, all fitted to the saturation properties of nuclear matter, as well. The omission or inclusion of the VC is not as important as it is for Fig 1 either, here. The only noticeable effect is a slight (few MeV) shift in the position of the critical point which is, however, within the uncertainties of the phenomenological properties of nuclear matter, i.e. mainly due the variations in $\sigma(n_0)$, L and T_c , cf. Tab. II.

The evolution of the in-medium chiral condensate in the vicinity of the nuclear-matter LG transition may have interesting consequences for heavy-ion collisions at beam energies of the HADES and future CBM experiments at FAIR/GSI. As seen in the left panel of Fig. 2 the $T = 0$ chiral condensate jumps discontinuously at the onset of self-bound nuclear matter with saturation density n_0 . In the (μ_B, T) phase diagram this discontinuity persists up the critical point of the LG transition, from where on it changes into a rather sharp crossover, which only gradually widens with decreasing μ_B , as seen in Fig. 4.

The chemical freeze-out points, extracted from Refs. [43, 44], are located along this crossover region,

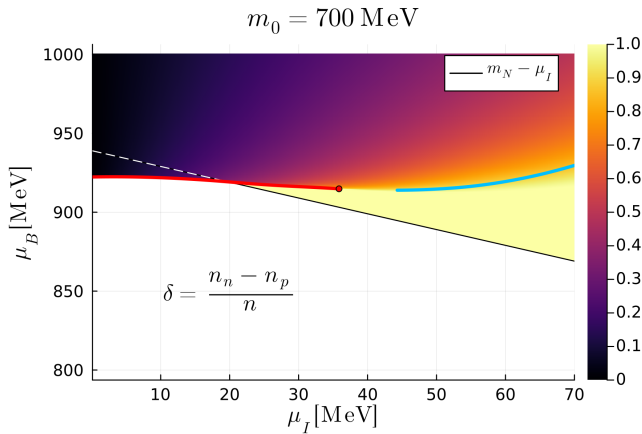


FIG. 3. The zero-temperature phase diagram in the (μ_I, μ_B) -plane. The heatmap displays the value of the isospin asymmetry coefficient $\delta = (n_n - n_p)/n$. The β -equilibrium line is highlighted in light blue.

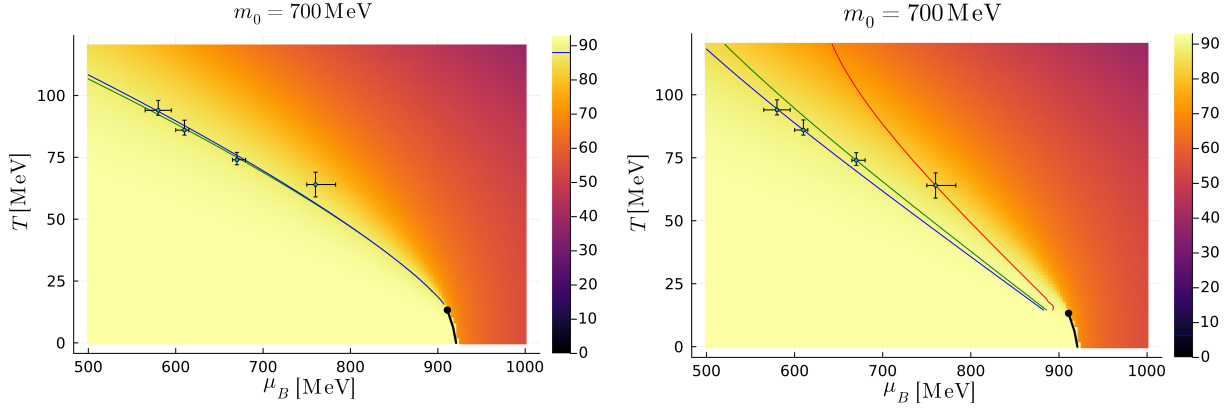


FIG. 4. Left panel: zoom-in of the (μ_B, T) phase diagram (for $m_0 = 700$ MeV and $\mu_I = 0$) around the LG transition. The heatmap displays values of the chiral condensate $\sigma(T, \mu_B)$. The green line represents the chemical freeze-out line at density $n = 0.15 n_0$ as discussed in Ref. [42] while the blue line denotes the line of a constant chiral condensate value of $\sigma = 88$ MeV. Right panel: lines of constant entropy/baryon, S/N of about 5.0 (red), 6.1 (green) and 6.3 (blue), passing through the chemical freeze-out points extracted in Ref. [43] [44]. Each line is associated to a different center-of-mass collision energy $\sqrt{s}$: red to 2.7 GeV, green to 3.3 GeV, and blue which passes through the points associated to both 3.8 GeV and 4.3 GeV.

and are well described by a line of constant density with $n = 0.15 n_0$ (green line in the left panel of Fig. 4) as previously noted in Ref. [42]. Interestingly, they are also described by a line of constant chiral condensate with a value of $\sigma = 88$ MeV (blue line in the left panel of Fig. 4). The fact that the contour lines of the density and the chiral condensate at their respective values emerging from the critical point stay so close to one another is an indication of a rather sharp crossover in this region of the phase diagram which also intuitively explains the proximity of the freeze-out points. This coincidence of the contour lines will not persist towards much smaller values of μ_B .

In the PDM, sizable modifications of $\sigma(T, \mu_B)$ directly translate into a reduction of the vacuum mass splitting, $\Delta m_{\pm}(f_{\pi}) = 571$ MeV, between the $N(939)$ and the $N^*(1535)$. For $T = 0$ and $\sigma(n_0) \simeq 68$ MeV at saturation, for example, one obtains $\Delta m_{\pm} \simeq 422$ MeV, and $\Delta m_{\pm} \rightarrow 0$ with the gradual restoration of chiral symmetry. With the value of $\sigma \simeq 88$ MeV in the vicinity of the chemical freeze-out line in the left panel of Fig. 4 it already reduces to $\Delta m_{\pm} \simeq 541$ MeV and hence falls below the mass of the η meson so that the $N^*(1535) \leftrightarrow \eta N$ channel closes. This may have observable implications for dilepton invariant-mass distributions at HADES/CBM energies [25, 48], dilepton angular distributions [49] and the electrical conductivity of nuclear matter [50].

B. Beta-Equilibrated Matter and Neutron Stars

In this section we discuss the implications of the thermodynamics of the PDM for isolated NS and NS mergers, focusing on the role of the in-medium chiral condensate for the EoS.

Since NS are charge neutral and in electro-weak equi-

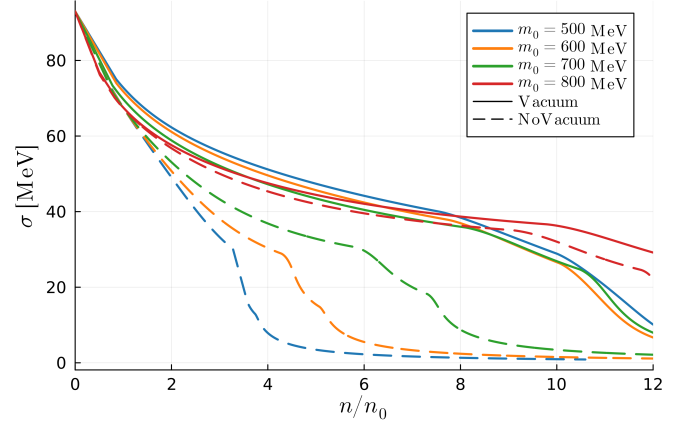


FIG. 5. The chiral condensate in β -equilibrated NS matter at $T = 0$ as a function of n/n_0 for various m_0 values, without (dashed lines) and with the VC (full lines). While for $m_0 = 800$ MeV the VC only induces small modifications, as before, for the lower values of m_0 the behavior of σ field changes significantly when the VC is neglected.

librium, the following conditions have to be imposed

$$\mu_n - \mu_p = \mu_e = \mu_{\mu}, \quad (44)$$

$$n_p = n_e + n_{\mu}. \quad (45)$$

where the subscripts p and n denote the iso-doublet of protons and neutrons together with their respective parity partners (p^* and n^*) while e and μ denote electrons and muons (it is assumed that the corresponding neutrinos, ν_e and ν_{μ} , freely escape during the NS evolution).

For zero-temperature neutron-star matter these conditions fix a line in the (μ_I, μ_B) -plane which is represented by the light blue line in Fig. 3. It begins in a region of high isospin asymmetry δ , at μ_I values beyond the critical point of the LG transition and moves continuously

through regions of lower δ due to the appearance of p , e and μ with increasing $\mu_I = (\mu_n - \mu_p)/2$.⁸

In Fig. 5 we display the chiral condensate as a function of baryon density in β -equilibrated NS matter. The chiral condensate evolves in the low density part as expected for pure neutron matter. The main difference is in the chiral transition that is triggered by the onset of the parity partner of the neutron, the n^* , at first. With increasing p^* fraction, eventually another drop is visible leading to a splitting of the chiral crossover. The splitting structure is more visible without the VC. It is washed out with the VC leaving a soft crossover that extends over a wide range of densities (roughly between $8n_0$ and $10n_0$) and makes the evolution of the condensate less sensitive to m_0 (at least between 500 MeV and 700 MeV).

After the abundant evidence of the importance of the VC, we focus exclusively on the parameter sets with the VC included, in the discussion of the cold NS properties from the PDM for the rest of this section.

In Fig. 6 the relative abundances of neutrons and protons as well as their parity partners, n^* and p^* , together with electrons and muons are shown for zero-temperature β -equilibrated matter with $m_0 = 500$ MeV (left panel) and $m_0 = 800$ MeV (right panel). While n is the baryon density as before, with n_{tot} we here denote the total particle-number density. Since we have assumed proton and neutron isospin degeneracy, there is no threshold for the appearance of electrons.

Finally, we discuss the EoS of cold NS matter. Fig. 7 displays the pressure P as a function of energy density ϵ (left panel), and the squared speed of sound c_s^2 as function of density (right panel). In c_s^2 the effects of the chiral transition manifest themselves as cusps and troughs reflecting the onset of the n^* and p^* degrees of freedom.

The speed of sound exceeds the conformal limit of asymptotically free quark matter. This is not unexpected for an effective baryonic model with short-range vector repulsion, however. In fact, with $\bar{\omega} \propto n$ for large μ_B the energy associated with the ω expectation value is not negligible compared to μ_B , and one necessarily obtains $c_s^2 \rightarrow 1$ while the conformal limit requires the chemical potential to be the only relevant energy scale for $n \rightarrow \infty$. In principle, the expected conformal limit with $c_s^2 \rightarrow 1/3$ could be restored by adding quartic self-interactions $\propto \omega^4$ [54, 55]. We do not apply this fix here because the short-distance repulsion between the baryons as an effective QCD interaction is eventually expected to be dissolved with asymptotic freedom in the limit of large μ_B . For a recent discussion of the speed of sound and its approach to the conformal limit see Ref. [56].

⁸ The phase diagram is calculated fixing a pair (μ_I, μ_B) and computing the corresponding n and n_I that minimize the grand potential. This procedure, actually, prevents the exploration of the system at certain baryon and isospin densities. For this reason we report the β -equilibrium line in Fig. 3 only where it is compatible with the underlying phase diagram.

No Vacuum Contribution

m_0 [MeV]	$M_{\text{max}} [M_\odot]$	$\epsilon_{\text{max}} [\text{MeV}/\text{fm}^3]$	$n_{\text{max}} [n_0]$
500	1.94	1479	7.42
600	1.87	1395	7.19
700	1.76	1383	7.22
800	1.53	1576	8.26

Vacuum Contribution

m_0 [MeV]	$M_{\text{max}} [M_\odot]$	$\epsilon_{\text{max}} [\text{MeV}/\text{fm}^3]$	$n_{\text{max}} [n_0]$
500	1.84	1383	7.05
600	1.77	1527	7.7
700	1.69	1525	7.79
800	1.52	1617	8.43

TABLE III. Maximum mass, as well as the energy-density and the number density in the NS center for the various values of m_0 considered here, without (top) and with the VC (bottom) for comparison.

Comparing with the values in Tab. III we can see that for each m_0 the chiral transition happens after the maximum density supported by the neutron star. This implies that in the two-flavor PDM we encounter no chirally restored matter in cold neutron stars once the VC is taken into account. The possibility of chirally restored matter in proto-neutron stars or in NS post-mergers remain open.

Fig. 8 reports, in the left panel, the M-R relations resulting from our EoS's matched to the HS(DD2) EoS [57, 58] for the low density part up to n_0 . The solution of TOV equation is provided by the software toolkit TOVExtravaganza [59]. As we can see from Fig. 8 and Tab. III, the maximum masses are below 2 solar masses for each parameter set. This implies that the present model is not able to reproduce the astrophysical constraint by the mass of PSR J0740+6620 [52], with the smaller values of m_0 coming closer to this constraint than the larger ones. The right panel of Fig. 8 shows the dimensionless tidal deformability Λ_{tidal} as a function of the stellar mass. Here, we find agreement with current gravitational-wave observations [53] for all values of m_0 .

Another key quantity of interest for NS matter, derived from the EoS, is the thermal index,

$$\Gamma_{\text{th}} = 1 + \frac{P_{\text{th}}}{\epsilon_{\text{th}}}, \quad (46)$$

with

$$P_{\text{th}} = P(\mu_B, \delta, T) - P(\mu_B, \delta, 0), \quad (47)$$

$$\epsilon_{\text{th}} = \epsilon(\mu_B, \delta, T) - \epsilon(\mu_B, \delta, 0). \quad (48)$$

It encodes the finite-temperature effects on the EoS, which are crucial to understand the role of temperature in proto-neutron stars and binary NS merger events. Generally, the thermal index shows a strong dependence on density and particle composition. A higher thermal index can lead to greater ejecta masses for a stiff EoS but can decrease for a softer EoS. In our case the softening is induced by the population of the parity-partner degrees of freedom.

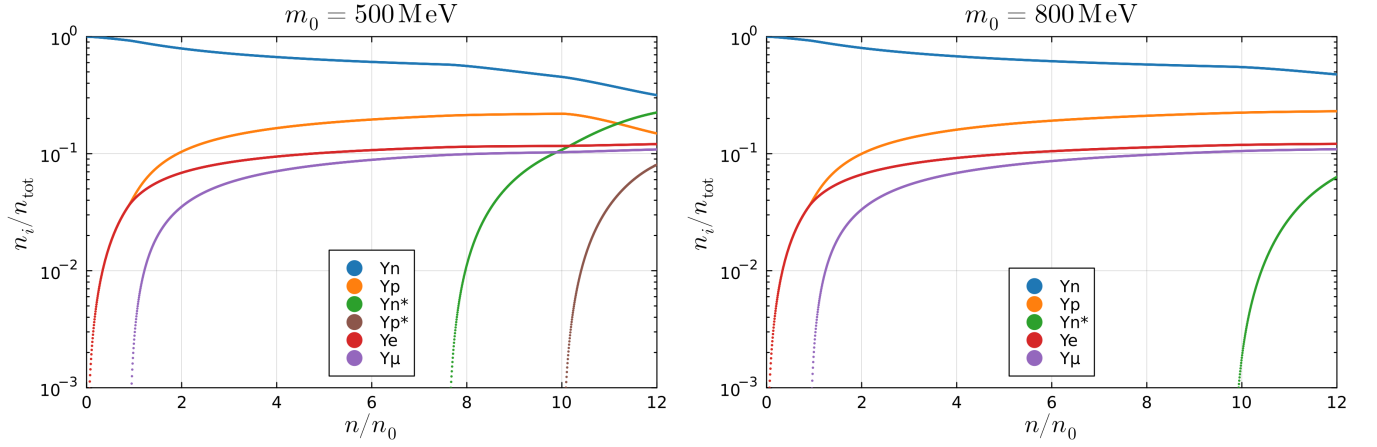


FIG. 6. Relative abundances of the various species in cold β -equilibrated NS matter as a function of baryon density n/n_0 .

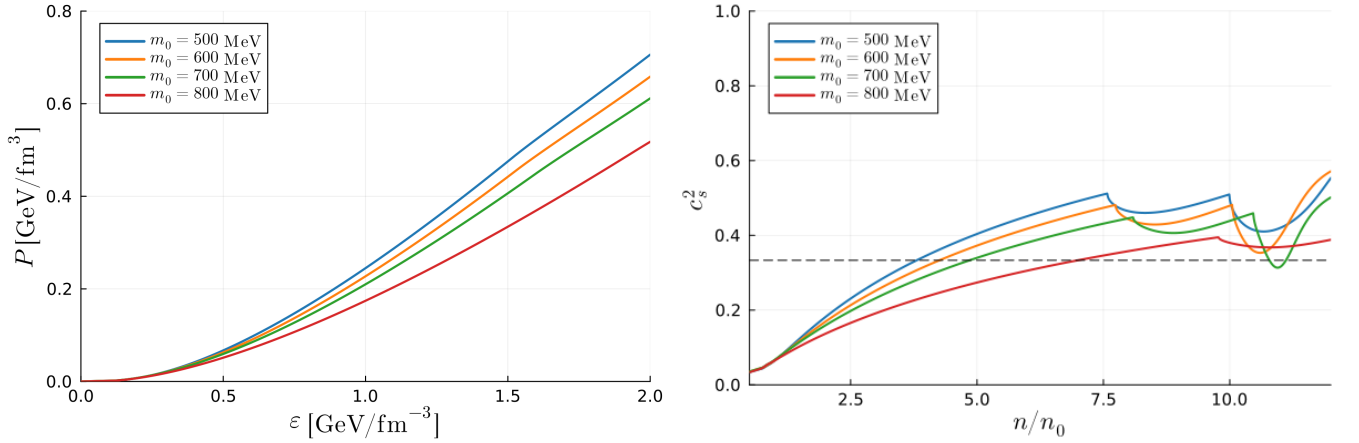


FIG. 7. The EoS (left panel) and the squared speed of sound (right panel) of cold NS matter for the various values of m_0 . The dashed line in the right panel denotes the conformal limit $c_s^2 = 1/3$.

The results shown in Fig. 9, are similar to what is obtained for the onset of hyperons in an $SU(3)$ EoS where a strong density dependence is observed [60]. The same is found in our case, albeit at much higher densities. The onset of the parity partners n^* and p^* (see Fig. 6) above $n/n_0 \sim 8$ induces rapid variations in Γ_{th} which are most pronounced at low temperatures.

In the low-density limit, the presence of electrons, which are ultra-relativistic at every temperature reported here, bring Γ_{th} close to the ultra-relativistic limit of $4/3$.

IV. CONCLUSIONS AND OUTLOOK

In this work we have introduced a multiplicatively renormalizable MF formulation of the two-flavor PDM in which the grand-canonical potential Ω is explicitly expressed in an RG-invariant form. This RG-MF calculation is based on a parametrization of the mesonic effective potential $V(\phi^2)$, polynomial in $\phi^2 = \sigma^2 + \vec{\pi}^2$, for sigma meson and pions in terms of suitable powers of the parity-

doublet baryon masses $m_{\pm}(\phi)$. In this formulation, the renormalization scale and the relevant renormalized dimensionless coupling are replaced by the RG-invariant scale parameter Λ of the PDM, in a scheme where the reference scale in the renormalization condition is set to the chirally invariant PDM baryon mass m_0 .

For applications of the model to dense baryonic matter we have fitted (for a given m_0) the strengths of isoscalar g_ω and isovector g_ρ vector repulsion between the baryons, and two dimensionless higher-order couplings c_6 and c_8 in our parametrization of the polynomial $V(\phi^2)$ (for the degree $n = 6$ and 8 interactions in units of m_0) to the properties of nuclear matter around saturation: the saturation density n_0 , the binding energy per nucleon E_B , the nuclear compressibility K_∞ , the symmetry energy E_{sym} and the in-medium chiral condensate $\sigma(n_0)$. The values for the slope parameter L and the critical temperature T_c of the nuclear LG transition resulting from the model calculation were not fitted but used as important sanity checks. Using this input from nuclear physics we have calculated chiral phase diagrams for symmetric and

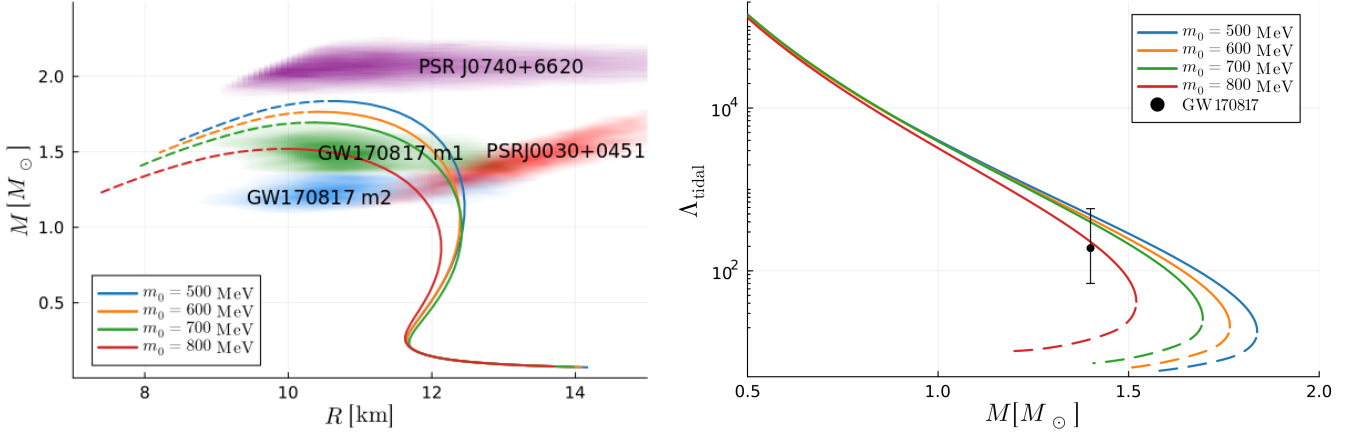


FIG. 8. Left panel: The M-R relation for evolved NS, obtained with the parameter sets discussed in the text. The shaded colored areas indicate constraints from astrophysical observations given in [51], [52] and [53]. Right panel: The tidal deformability as a function of the stellar mass for the various values of m_0 . All cases are consistent with the 90% CL value of the current gravitational-wave observation [53].

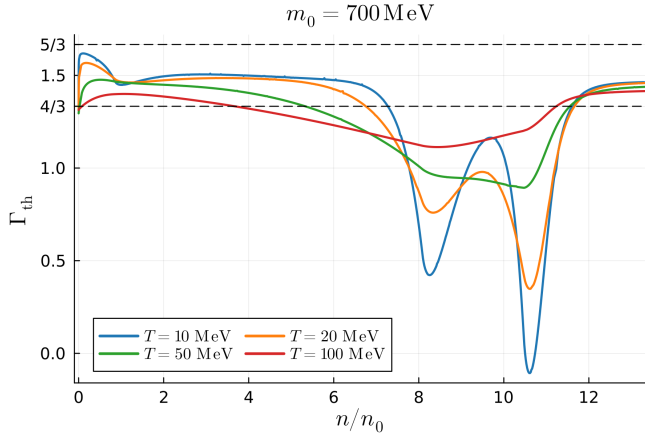


FIG. 9. The thermal index Γ_{th} as a function of density in β -equilibrated NS matter for various temperatures, here exemplarily with $m_0 = 700$ MeV. The horizontal lines indicate the free Fermi gas limit for Γ_{th} : $5/3$ for the non-relativistic gas and $4/3$ for the ultra-relativistic one.

asymmetric nuclear matter for a wide range of temperatures as well as baryo- and isospin chemical potentials.

A major finding is that the inclusion of the fermionic VC to Ω moves the chiral transition to larger baryo-chemical potentials μ_B for symmetric nuclear matter or μ_n for pure neutron matter while substantially weakening any rapid low-temperature variations, especially for smaller values of m_0 . At the same time, the position of the LG critical point as well as the zero-temperature phase diagram in the (μ_B, μ_I) -plane are essentially unaffected by the choice of m_0 .

For β -equilibrated cold neutron-star matter, we find that chiral restoration, and the relative appearance of the negative-parity partners of the neutron and proton, n^* and p^* , lead to cusps and troughs in c_s^2 and rapid

variations in the low-temperature thermal index Γ_{th} . As these occur at densities beyond $7n_0$ which, in all cases studied, are above the central density n_{max} of the star and thus irrelevant for the M-R relation. The values for the maximum mass tend to undershoot the observational $M_{\text{max}} \geq 2M_\odot$ constraints, especially for larger values of m_0 . From this we conclude that a more detailed description of the nuclear hard-core repulsion is needed.

In view of our findings it is imperative to revisit the question of what the role of strangeness is and whether hyperons appear in the inner core of NS. We wish to address this issue in future work by extending the RG-consistent framework, discussed here, by promoting the PDM to the $SU(3)$ -symmetric case, thus incorporating kaons and hyperons as well as the $U(1)_A$ anomaly.

To assess modifications of the finite-temperature MF EoS it will be important to include fluctuations of the meson fields, especially the pion field, which can give sizable contributions to the thermal pressure, especially at high temperature [61]. A proper treatment of fluctuations of vector- and axial-vector fields around the RG-invariant MF will determine the electro-weak response and allow for quantitative predictions of dilepton and neutrino properties for heavy-ion experiments and NS observations [50]. Work in this direction is in progress.

ACKNOWLEDGMENTS

We thank Anton Andronic for discussions on the chemical freeze out at high-baryon densities and providing us with pertinent freeze-out data, Michael Buballa and Bengt Friman for discussions on the generic features of the zero-temperature phase diagram, Hosein Gholami on RG consistency and for help with his open-source software used to solve the TOV equations, Ugo Mire for discussions and exchanges, especially on the numer-

ical implementation of this work, and Jürgen Schaffner-Bielich for discussions on astrophysical constraints and effects of including strangeness. This work was supported by the Deutsche Forschungsgemeinschaft, project

no. 315477589, the Collaborative Research Center CRC-TR 211, “Strong-interaction matter under extreme conditions.”

-
- [1] S. S. Afonin, Parity Doubling in Particle Physics, *Int. J. Mod. Phys. A* **22**, 4537 (2007), arXiv:0704.1639 [hep-ph].
 - [2] C. E. Detar and T. Kunihiro, Linear σ Model With Parity Doubling, *Phys. Rev. D* **39**, 2805 (1989).
 - [3] M. Gell-Mann and M. Levy, The axial vector current in beta decay, *Nuovo Cim.* **16**, 705 (1960).
 - [4] B. D. Serot and J. D. Walecka, The Relativistic Nuclear Many Body Problem, *Adv. Nucl. Phys.* **16**, 1 (1986).
 - [5] C. E. Detar and J. B. Kogut, The Hadronic Spectrum of the Quark Plasma, *Phys. Rev. Lett.* **59**, 399 (1987).
 - [6] C. E. Detar and J. B. Kogut, Measuring the Hadronic Spectrum of the Quark Plasma, *Phys. Rev. D* **36**, 2828 (1987).
 - [7] G. Aarts, C. Allton, D. De Boni, S. Hands, B. Jäger, C. Praki, and J.-I. Skullerud, Light baryons below and above the deconfinement transition: medium effects and parity doubling, *JHEP* **06**, 034, arXiv:1703.09246 [hep-lat].
 - [8] M. Marczenko, D. Blaschke, K. Redlich, and C. Sasaki, Parity Doubling and the Dense Matter Phase Diagram under Constraints from Multi-Messenger Astronomy, *Universe* **5**, 180 (2019), arXiv:1905.04974 [nucl-th].
 - [9] Y. K. Kong, T. Minamikawa, and M. Harada, Neutron star matter based on a parity doublet model including the $a_0(980)$ meson, *Phys. Rev. C* **108**, 055206 (2023), arXiv:2306.08140 [nucl-th].
 - [10] J. Eser and J.-P. Blaizot, Thermodynamics of the parity-doublet model: Symmetric nuclear matter and the chiral transition, *Phys. Rev. C* **109**, 045201 (2024), arXiv:2309.06566 [nucl-th].
 - [11] J. Steinheimer, S. Schramm, and H. Stocker, The hadronic SU(3) Parity Doublet Model for Dense Matter, its extension to quarks and the strange equation of state, *Phys. Rev. C* **84**, 045208 (2011), arXiv:1108.2596 [hep-ph].
 - [12] T. Minamikawa, B. Gao, T. Kojo, and M. Harada, Chiral Restoration of Nucleons in Neutron Star Matter: Studies Based on a Parity Doublet Model, *Symmetry* **15**, 745 (2023), arXiv:2302.00825 [nucl-th].
 - [13] E. S. Fraga, R. da Mata, and J. Schaffner-Bielich, SU(3) parity doubling in cold neutron star matter, *Phys. Rev. D* **108**, 116003 (2023), arXiv:2309.02368 [hep-ph].
 - [14] P. Senger (CBM), Probing dense QCD matter in the laboratory—The CBM experiment at FAIR, *Phys. Scripta* **95**, 074003 (2020), arXiv:2005.03321 [nucl-ex].
 - [15] J. Adamczewski-Musch *et al.* (HADES), HADES and the QCD phase diagram, *PoS CPOD2021*, 003 (2022).
 - [16] J. Eser and J.-P. Blaizot, Thermodynamics of the parity-doublet model. II. Asymmetric and neutron matter, *Phys. Rev. C* **110**, 065205 (2024), arXiv:2408.01302 [nucl-th].
 - [17] B. B. Brandt, V. Chelnokov, G. Endrodi, G. Marko, D. Scheid, and L. von Smekal, Renormalization group invariant mean-field model for QCD at finite isospin density, *Phys. Rev. D* **112**, 054038 (2025), arXiv:2502.04025 [hep-ph].
 - [18] D. Jido, Y. Nemoto, M. Oka, and A. Hosaka, Chiral symmetry for positive and negative parity nucleons, *Nucl. Phys. A* **671**, 471 (2000), arXiv:hep-ph/9805306.
 - [19] Y.-B. Yang, J. Liang, Y.-J. Bi, Y. Chen, T. Draper, K.-F. Liu, and Z. Liu, Proton Mass Decomposition from the QCD Energy Momentum Tensor, *Phys. Rev. Lett.* **121**, 212001 (2018), arXiv:1808.08677 [hep-lat].
 - [20] V. D. Burkert, L. Elouadrhiri, F. X. Girod, C. Lorcé, P. Schweitzer, and P. E. Shanahan, Colloquium: Gravitational form factors of the proton, *Rev. Mod. Phys.* **95**, 041002 (2023), arXiv:2303.08347 [hep-ph].
 - [21] S. Navas *et al.* (Particle Data Group), Review of particle physics, *Phys. Rev. D* **110**, 030001 (2024).
 - [22] V. Skokov, B. Friman, E. Nakano, K. Redlich, and B. J. Schaefer, Vacuum fluctuations and the thermodynamics of chiral models, *Phys. Rev. D* **82**, 034029 (2010), arXiv:1005.3166 [hep-ph].
 - [23] L. Brandes, N. Kaiser, and W. Weise, Fluctuations and phases in baryonic matter, *Eur. Phys. J. A* **57**, 243 (2021), arXiv:2103.06096 [nucl-th].
 - [24] J. Weyrich, N. Strodthoff, and L. von Smekal, Chiral mirror-baryon-meson model and nuclear matter beyond mean-field approximation, *Phys. Rev. C* **92**, 015214 (2015), arXiv:1504.02697 [nucl-th].
 - [25] A. B. Larionov and L. von Smekal, Effects of chiral symmetry restoration on meson and dilepton production in relativistic heavy-ion collisions, *Phys. Rev. C* **105**, 034914 (2022), arXiv:2109.03556 [nucl-th].
 - [26] J. Kim and S. H. Lee, Masses of hadrons in the chiral symmetry restored vacuum, *Phys. Rev. D* **105**, 014014 (2022), arXiv:2109.12791 [hep-ph].
 - [27] C. Drischler, P. G. Giuliani, S. Bezoui, J. Piekarewicz, and F. Viens, Bayesian mixture model approach to quantifying the empirical nuclear saturation point, *Phys. Rev. C* **110**, 044320 (2024), arXiv:2405.02748 [nucl-th].
 - [28] J. R. Stone, N. J. Stone, and S. A. Moszkowski, Incompressibility in finite nuclei and nuclear matter, *Phys. Rev. C* **89**, 044316 (2014), arXiv:1404.0744 [nucl-th].
 - [29] M. Baldo and G. F. Burgio, The nuclear symmetry energy, *Prog. Part. Nucl. Phys.* **91**, 203 (2016), arXiv:1606.08838 [nucl-th].
 - [30] A. Agadjanov, D. Djukanovic, G. von Hippel, H. B. Meyer, K. Ottnad, and H. Wittig, Nucleon Sigma Terms with $N_f=2+1$ Flavors of O(a)-Improved Wilson Fermions, *Phys. Rev. Lett.* **131**, 261902 (2023), arXiv:2303.08741 [hep-lat].
 - [31] C. Alexandrou, S. Bacchio, J. Finkenrath, C. Iona, G. Koutsou, Y. Li, and G. Spanoudes, Nucleon charges and σ -terms in lattice QCD, *Phys. Rev. D* **111**, 054505 (2025), arXiv:2412.01535 [hep-lat].
 - [32] M. Hoferichter, J. Ruiz de Elvira, B. Kubis, and U.-G. Meißner, High-Precision Determination of the Pion-Nucleon σ Term from Roy-Steiner Equations, *Phys. Rev. Lett.* **115**, 092301 (2015), arXiv:1506.04142 [hep-ph].

- [33] M. Hoferichter, J. R. de Elvira, B. Kubis, and U.-G. Meißner, On the role of isospin violation in the pion–nucleon σ -term, *Phys. Lett. B* **843**, 138001 (2023), arXiv:2305.07045 [hep-ph].
- [34] S. Owa, D. B. Leinweber, A. W. Thomas, and X.-G. Wang, Chiral analysis of the nucleon mass and sigma commutator, *Phys. Rev. D* **109**, 116022 (2024), arXiv:2310.12564 [hep-ph].
- [35] N. Kaiser, P. de Homont, and W. Weise, In-medium chiral condensate beyond linear density approximation, *Phys. Rev. C* **77**, 025204 (2008), arXiv:0711.3154 [nucl-th].
- [36] V. A. Karnaukhov and A. S. Botvina, Comment on ‘The experimental liquid-vapor phase diagram of bulk nuclear matter’, *J. Phys. G* **40**, 058001 (2013).
- [37] B. K. Agrawal, J. N. De, and S. K. Samaddar, Determining the density content of symmetry energy and neutron skin: an empirical approach, *Phys. Rev. Lett.* **109**, 262501 (2012), arXiv:1212.0292 [nucl-th].
- [38] B. K. Agrawal, J. N. De, S. K. Samaddar, G. Colo, and A. Sulaksono, Constraining the density dependence of the symmetry energy from nuclear masses, *Phys. Rev. C* **87**, 051306 (2013), arXiv:1305.5336 [nucl-th].
- [39] H. Gil, Y.-M. Kim, P. Papakonstantinou, and C. H. Hyun, Constraining the density dependence of the symmetry energy with nuclear data and astronomical observations in the Korea-IBS-Daegu-SKKU framework, *Phys. Rev. C* **103**, 034330 (2021), arXiv:2010.13354 [nucl-th].
- [40] B.-A. Li, B.-J. Cai, W.-J. Xie, and N.-B. Zhang, Progress in Constraining Nuclear Symmetry Energy Using Neutron Star Observables Since GW170817, *Universe* **7**, 182 (2021), arXiv:2105.04629 [nucl-th].
- [41] F. J. Fattoyev, C. J. Horowitz, J. Piekarewicz, and G. Shen, Relativistic effective interaction for nuclei, giant resonances, and neutron stars, *Phys. Rev. C* **82**, 055803 (2010), arXiv:1008.3030 [nucl-th].
- [42] S. Floerchinger and C. Wetterich, Chemical freeze-out in heavy ion collisions at large baryon densities, *Nucl. Phys. A* **890-891**, 11 (2012), arXiv:1202.1671 [nucl-th].
- [43] A. Andronic, P. Braun-Munzinger, K. Redlich, and J. Stachel, Decoding the phase structure of QCD via particle production at high energy, *Nature* **561**, 321 (2018), arXiv:1710.09425 [nucl-th].
- [44] A. Andronic, P. Braun-Munzinger, B. Friman, P. M. Lo, K. Redlich, and J. Stachel, The thermal proton yield anomaly in Pb-Pb collisions at the LHC and its resolution, *Phys. Lett. B* **792**, 304 (2019), arXiv:1808.03102 [hep-ph].
- [45] W.-j. Fu, J. M. Pawłowski, and F. Rennecke, QCD phase structure at finite temperature and density, *Phys. Rev. D* **101**, 054032 (2020), arXiv:1909.02991 [hep-ph].
- [46] P. J. Gunkel and C. S. Fischer, Locating the critical endpoint of QCD: Mesonic backcoupling effects, *Phys. Rev. D* **104**, 054022 (2021), arXiv:2106.08356 [hep-ph].
- [47] K. Kamikado, N. Strodthoff, L. von Smekal, and J. Wambach, Fluctuations in the quark-meson model for QCD with isospin chemical potential, *Phys. Lett. B* **718**, 1044 (2013), arXiv:1207.0400 [hep-ph].
- [48] F. Geurts and R.-A. Tripolt, Electromagnetic probes: Theory and experiment, *Prog. Part. Nucl. Phys.* **128**, 104004 (2023), arXiv:2210.01622 [hep-ph].
- [49] F. Seck, B. Friman, T. Galatyuk, H. van Hees, R. Rapp, E. Speranza, and J. Wambach, Polarization of thermal dilepton radiation, *Phys. Lett. B* **861**, 139267 (2025), arXiv:2309.03189 [nucl-th].
- [50] J. Wambach, Baryon-rich QCD matter, *J. Phys. Conf. Ser.* **2536**, 012002 (2023), arXiv:2304.14833 [hep-ph].
- [51] M. C. Miller *et al.*, PSR J0030+0451 Mass and Radius from *NICER* Data and Implications for the Properties of Neutron Star Matter, *Astrophys. J. Lett.* **887**, L24 (2019), arXiv:1912.05705 [astro-ph.HE].
- [52] A. J. Dittmann *et al.*, A More Precise Measurement of the Radius of PSR J0740+6620 Using Updated *NICER* Data, *Astrophys. J.* **974**, 295 (2024), arXiv:2406.14467 [astro-ph.HE].
- [53] B. P. Abbott *et al.* (LIGO Scientific, Virgo), GW170817: Measurements of neutron star radii and equation of state, *Phys. Rev. Lett.* **121**, 161101 (2018), arXiv:1805.11581 [gr-qc].
- [54] H. Mueller and B. D. Serot, Relativistic mean field theory and the high density nuclear equation of state, *Nucl. Phys. A* **606**, 508 (1996), arXiv:nucl-th/9603037.
- [55] E. S. Fraga, R. da Mata, S. Pitsinikos, and A. Schmitt, Strange quark matter from a baryonic approach, *Phys. Rev. D* **106**, 074018 (2022), arXiv:2206.09219 [nucl-th].
- [56] K. Fukushima and S. Minato, Speed of sound and trace anomaly in a unified treatment of the two-color diquark superfluid, the pion-condensed high-isospin matter, and the 2SC quark matter, *Phys. Rev. D* **111**, 094006 (2025), arXiv:2411.03781 [hep-ph].
- [57] S. Typel, G. Röpke, T. Klähn, D. Blaschke, and H. H. Wolter, HS(DD2) Neutron Matter (with electrons), *Zenodo* (2018), data hosted via <https://compose.obspm.fr/eos/2>.
- [58] S. Typel, G. Röpke, T. Klähn, D. Blaschke, and H. H. Wolter, Composition and thermodynamics of nuclear matter with light clusters, *Phys. Rev. C* **81**, 015803 (2010), arXiv:0908.2344 [nucl-th].
- [59] H. Gholami, TOVExtravaganza: Python toolkit for solving the Tolman-Oppenheimer-Volkoff (TOV) equations and exploring neutron star properties (2025).
- [60] H. Kochankowski, G. Lioutas, S. Blacker, A. Bauswein, A. Ramos, and L. Tolos, Impact of hyperons on neutron star mergers: Gravitational waves, mass ejection, and black hole formation, *Phys. Rev. D* **112**, 023014 (2025), arXiv:2501.12905 [astro-ph.HE].
- [61] B. Fore and S. Reddy, Pions in hot dense matter and their astrophysical implications, *Phys. Rev. C* **101**, 035809 (2020), arXiv:1911.02632 [astro-ph.HE].